

How the X and Y Became the Sex Chromosomes

In 1960, an international body of geneticists assembled in Denver, Colorado, to agree on a system for naming and classifying the human chromosomes. The X and Y presented a point of debate and conceptual dissonance. The inherited system, devised in the first decades of the century, postulated two kinds of chromosomes: the “sex” chromosomes and the “ordinary” chromosomes, or autosomes. The scientists wondered: is there justification to continue this central distinction between sex chromosomes and the others? Or should the X and Y simply be classified by size and structure, like autosomes? Why are the X and Y called the “sex chromosomes,” after all?¹

The distinction between “ordinary” and “other” chromosomes was honed in the early twentieth century. Debates over cytological nomenclature raged on the pages of science journals. The question of the “chromosome for sex” took center stage. By 1910 the X and Y had “received a great variety of names,” wrote Thomas Montgomery.² The most common terms for the X and Y chromosomes, and their first traceable appearance in the early literature, include odd chromosomes (1890s), accessory chromosomes (1899), heterochromosomes (1904), idiochromosomes (1905), heterotropic chromosomes (1905), monosomes/diplosomes (1906), gonochromosomes (1906), sex chromosomes (1906), and supernumerary chromosomes (1908).³ Three terms, “accessory chromosomes,” “heterochromosomes,” and “idiochromosomes,” dominated literature on the

X and Y from the first hypothesis of a link between the X and sex until the 1920s, when “sex chromosome” began to come into general use.⁴

The story of how the X and Y ultimately became the “sex chromosomes” offers a window into how contextual and contingent factors shaped the conceptual terrain of genomic sex from its earliest days. “Sex chromosome” was neither the first nor the most preferred name for the X and Y. Several alternative terms for the X and Y, which did not point to their link to sex, flourished into the 1920s. Many prominent biologists openly denounced the concept of a “sex chromosome.” So vituperous was the climate that the geneticist Thomas Hunt Morgan called the concept of the sex chromosome a “travesty.” To understand why “sex chromosome” triumphed, despite strong objections, we must visit the opportunistic role of the concept of the sex chromosome in consolidating two great biological efforts of the first quarter of the twentieth century: the chromosomal theory of inheritance (this chapter) and later the new biology of sex heralded by the discovery of the “sex hormones” (chapter 4).

THE TERMINOLOGY OF ABERRANT CHROMOSOMES

Each of the popular terminological contenders for the X and Y—“accessory,” “heterochromosome,” “idiochromosome,” and “sex chromosome”—captured different features of the X and Y and reflected a different view of the empirical and theoretical significance of these unusual chromosomes.

Coined in 1899 by Clarence McClung, the term “accessory chromosome” originates in observations of “XO” insect species, in which half of the sperm appeared to have an extra, or accessory, chromosome. The term was first applied to the X chromosome. It would later be used generically to refer to the X and Y—as in the “accessory elements.”⁵ “Accessory chromosome” was used far more widely than “sex chromosome” and remained popular into the 1920s. McClung insisted on the superiority of “accessory chromosome” to “competitive terms with improper associations.”⁶ Compared to the alternatives, the term “accessory chromosome” laid special emphasis on the existence of an “extra” chromosome in one sex.

Montgomery’s “heterochromosome” joined the lexicon in 1904. Initially, Montgomery intended the prefix *hetero-* to distinguish the X, which he believed was a nonautonomous, “reformulated” chromosome, from the ordinary *auto-*somes, which were independent cell bodies that maintain their form. But “heterochromosome” quickly became the pre-

ferred term for the X and Y for other reasons—it seemed to describe precisely their status as a heterogeneous pairing unit. Montgomery argued that the pairing behavior of chromosomes, not the number of chromosomes, should take center stage in chromosome naming. He pressed “the value of chromosomal relations as a taxonomic character” as opposed to the “less constant” “number of the chromosomes.”⁷ The status of the X as an “extra” or accessory chromosome was not as central, he argued, as its status as a nonpairing or hetero-pairing chromosome.

Nettie Stevens favored “heterochromosome” for the XX/XY type. The term “heterochromosome” offered the additional advantage, over the term “accessory chromosome,” of highlighting the commonly observed size differential between the X and Y. She first used it in 1905 to distinguish XY (heterochromosome) from XO (accessory chromosome) forms of sex determination. Describing the two sizes of chromosomes she found in the worm *Tenebrio*, for example, she wrote: “This seems to be a clear case of sex-determination, *not by an accessory chromosome*, but by a definite difference in the character of the elements of one pair of chromosomes.”⁸ Rather than an accessory chromosome, Stevens described *Tenebrio* as possessing “smaller” and “larger” heterochromosomes.⁹

Wilson proposed an alternative term, “idiochromosome,” in 1905 to describe cases of two paired unequal-sized chromosomes. He called the X the “heterotropic chromosome” and saw the X as one of a *pair*, either present or ancestral, of sex-determining chromosomes. Wilson specifically disfavored the term “accessory chromosome.” Defending his alternative, Wilson wrote:

Since there is no reason for considering the “accessory chromosome” as in any sense accessory to the others, it appears to me that McClung’s term might well be abandoned in favor of a less compromising one. I suggest that until their physiological significance is positively determined chromosomes of this type may provisionally be called *heterotropic* chromosomes (in allusion to the fact that they pass to one pole only of the spindle in one of the maturation-divisions) in contradistinction to *amphitropic* chromosomes, the products of which pass to both poles in both divisions.¹⁰

Wilson’s system of nomenclature stressed chromosome pairing behavior during cell division. Males had a pair of idiochromosomes, large and small (XY); females had a single heterotropic chromosome (XO).

The term “idiochromosome” emphasized the sportive, nonhomologous features of the X and Y chromosomes and, like Montgomery’s term,

focused on the fact that the X and Y are a pair of “unequal” chromosomes. As Wilson defined them, idiochromosomes are those “left without a fellow of like size.”¹¹ “The *purely descriptive* term ‘idiochromosome’ (peculiar or distinctive chromosomes),” wrote Wilson, “will be applied to the two chromosomes, usually unequal in size, which . . . undergo a very late conjugation and subsequent asymmetrical distribution to the spermatid-nuclei.”¹² It was the distinctive, idiosyncratic pairing behavior of the X and Y, not their link to sex, that merited a special term for them. As Wilson put it, “were it not for their failure to unite to form a bivalent body until the end of the first mitosis we should find *no ground* in this case for designating these chromosomes by a special name.”¹³

In a 1906 contribution to *Science* titled, in part, “The Terminology of Aberrant Chromosomes,” Montgomery, who had recently coined the term “autosome” for the “ordinary chromosomes,” decried the proliferation of nomenclature for the X and Y and asserted “a pressing need for a conciser and more uniform nomenclature.” “I call upon fellow workers to discard their previous names,” Montgomery enjoined.¹⁴ While “autosome” quickly became the term of choice for homologous chromosomes, the question of what to call the X and Y would remain unsettled. In a 1913 *Science* editorial, McClung, a stickler for terminology who had now been elevated to the chair of zoology at the University of Pennsylvania, charged that cytological nomenclature had “fallen into lamentable confusion.” This, he argued, was due to the scattershot naming of odd chromosomes, including the X and Y, by a *function* or *character* rather than by an accurate pointer to physical identification.¹⁵

The term “sex chromosome” was indeed inconsistent with the predominant descriptive practices in chromosome studies. A demanding ocular art with an elusive and phasic object of study, chromosome science in its early days required a precise and rich descriptive convention. Chromosomes were named, characterized, and classified strictly by pairing behavior during meiosis, and by their size, shape, and structure. The term “sex chromosome” transgressed this classification system. It specified the chromosome instead by the *trait* putatively linked to it. As such, researchers worried that the concept of a sex chromosome might introduce distortions into reasoning about the relationship between chromosomes and traits. Specifically, the term “sex chromosome” raised suspicion of confused and overly deterministic reasoning about the relation between chromosomes and sex. Sensitive to this, even Wilson, the originator and eventual champion of the term, warned that it was meant only as a form of shorthand: “sex chromosome” was a designation, he wrote, to be used “whenever it is desirable to avoid circumlocution.”¹⁶

IN THE CROSSHAIRS OF MENDELISM

Below the surface of these terminological debates over the sex chromosome roiled a deep philosophical rift in turn-of-the-century biology. The sex chromosome arrived just as Mendel's new genetics was transforming and polarizing the world of biology. It became a flashpoint in these debates.

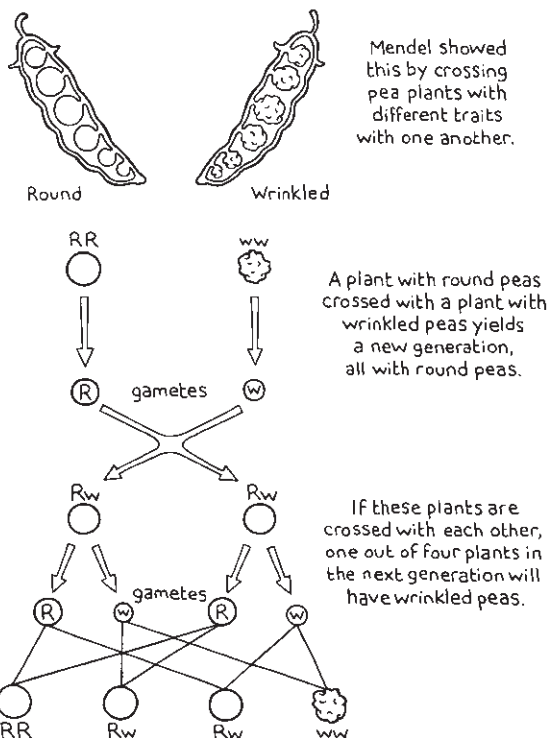
In 1900, three investigators rediscovered Mendel's laws of inheritance, which had been lying in obscurity since his plant hybrid experiments were first published in 1866.¹⁷ Mendel crossed plants that differed in discrete characters, such as his famous round and wrinkle-seeded peas. These experiments yielded a large body of data about the frequency of trait inheritance. This data showed that traits must be determined by pairs of factors, which today we call gene variants, or alleles, and that these factors can influence phenotype differently depending on whether they are dominant or recessive. Mendel's findings also predicted that these gene pairs segregate from one another independently during gamete formation and randomly recombine to produce genetic diversity. These results are known as Mendel's laws of *segregation* and *independent assortment* (see fig. 3.1).

By 1903, the British geneticist William Bateson had confirmed Mendel's laws of heredity, as well as the ideas of dominant and recessive traits and recombination. Mendelian genetics operated on simple, abstract quantitative principles. The research program could theoretically proceed without regard to the nature, location, or mechanism of the genetic material itself. Nonetheless a structure that accounted for the physical basis of Mendelism would provide the definitive evidence needed to establish Mendelism more broadly in biology. In this way, Mendelism became intimately tied up with the science of chromosomes already in progress.

In the first years of the twentieth century, German Theodor Boveri and American Walter Sutton demonstrated that chromosomes were autonomous and distinct bodies—good candidates to be the carriers of the material that assures genetic continuity. Boveri showed that each species is characterized by a distinctive chromosome set, that each chromosome is a discrete individual, and that homologous chromosomes of maternal and paternal origin pair during cell division. On this basis, Boveri believed that chromosomes specialized in different functions, carrying genetic factors for biological traits.¹⁸ Sutton asserted, furthermore, that chromosomes were the carriers of Mendelian alleles. Chromosomes come in homologous pairs—one inherited from each parent—that segregate and assort independently during cell division. Sutton explicitly con-

CHROMOSOMES PROVIDE A PHYSICAL BASIS FOR THE MENDELIAN LAWS OF INHERITANCE

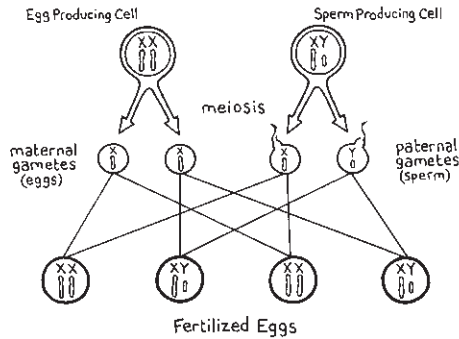
Mendel showed that traits are inherited in pairs of factors, one from each parent. The factors do not blend when inherited by offspring. Instead they segregate and re-assort when inherited to form new combinations.



nected the meiotic division and recombination of the chromosomes to the segregation of Mendelian hereditary factors. Over the next two decades, what we know today as the chromosomal theory of heredity took shape. The so-called sex chromosome would become the controversial vehicle of the chromosome theory and the new Mendelism.

Very early on, Mendelism became associated both with the X and Y chromosomes and with a conception of sex as an either-or binary. Bateson, the principal figure in the reinvigoration of Mendelism at the turn of the century, argued in 1900 that sex is a binary, discontinuous variation in nature and thus a prime candidate for Mendelian explanation.¹⁹

Sutton and Boveri showed that when a new cell is fertilized, each offspring receives one of each chromosome from its mother and its father. The chromosomes retain their physical autonomy.



In the case of the X chromosome, this feature was particularly clear to scientists.

The X chromosome could be clearly distinguished from the other chromosomes. It maintained its individuality during cell division and gamete formation.

Linked to trait of sex, the X chromosome also demonstrated that chromosomes can specialize in genetic traits.

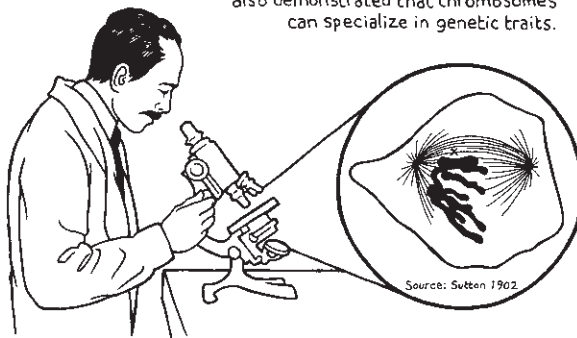


Figure 3.1. The chromosomal theory of inheritance. Illustration by Kendal Tull-Esterbrook; copyright Sarah S. Richardson.

A year following McClung's 1902 accessory chromosome hypothesis, Harvard geneticist William Castle, "the chief advocate of Mendelism in the United States,"²⁰ suggested that the X chromosome could provide the physical basis for a fully Mendelian genetic theory of sex and offered an elaborate model of how it might work.²¹

These Mendelian-chromosomal models of sex determination produced enormous irritation among cytologists and embryologists who were skeptical of Mendelism's broad claims about the nature of hered-

ity. They vigorously rejected the plausibility of a Mendelian account of sex, and along with it, the notion of a “sex chromosome.” They used the so-called sex chromosome to tar the Mendelian view that sex was a “unit character” as naïve and deterministic. French neo-Lamarckians such as Maurice Caullery joined British biometricians such as Karl Pearson and other skeptics to raise a chorus of criticism against the “sex chromosome.” In America, the leading critics included Thomas Montgomery and Thomas Hunt Morgan.

A chromosome expert at the University of Pennsylvania, Montgomery had earned his PhD in Berlin and was deeply influenced by the German cytological school. Montgomery’s opposition to the chromosomal theory of sex was philosophically grounded in his anti-Mendelism and his skepticism about Mendelism’s corollary, the chromosome theory of heredity. He derisively cited the notion of the sex chromosome as proof that “modern Mendelian explanations represent a determinant theory far more rigid and complex than that of Weismann” (who held that hereditary characters were transmitted by the germ plasm).²²

To Montgomery, the sex chromosome hypothesis represented an absurd and simplistic overextension of the chromosome theory of heredity. He maintained that the evidence showed only that the X chromosome and sex were “coincident,” not causally connected.²³ As he wrote in 1910, “there is no valid reason to interpret sex as an immutable unit character resident in or presided over by particular chromosomes, and sorted out and distributed by Mendelian segregation.”²⁴ He continued:

This nuclear element is not an independent unit, but only a part, even if it be the most important part, of the cell whole. Thus the idea is erroneous to speak of the chromosomes as automatic units, for they are but parts of the cell or cell complex. . . . Now to assume that particular chromosomes alone are sex-determinants is to disregard this complex inter-activity.²⁵

Moreover, the chromosomal theory of sex seemed to imply that maleness and femaleness are distinct, alternate developmental routes. In contrast, Montgomery argued that sex is “a labile process which may be changed by a variety of influences” and that maleness and femaleness are but “two modes of one process . . . not radically different conditions.”²⁶

Mendelism, critics such as Montgomery charged, required shoe-horning complex traits into discrete unit characters. The chromosomal theory of sex appeared symptomatic of this central anti-intellectualism of Mendelism. Wrote Montgomery:

He would be rash who would venture to claim that a particular chromosome determines excretion, another determines locomotion; yet these processes are relatively simple compared with that of sexuality, which some have contended may be controlled by a particular chromosome. The hypothesis is too naïve, it assumes too great simplicity of the cell, it tastes too strong of rigid predetermination.²⁷

The chromosome theory of sex, Montgomery believed, showed the true colors of the naïvely atomistic and deterministic view of heredity promoted by the Mendelians.²⁸

Morgan's condemnation of the "sex chromosome" was even more heated than Montgomery's. At the turn of the century, the young Morgan was emerging as a leading figure in American embryology. A 1900 trip abroad occasioned a meeting with the Dutch botanist and experimental geneticist Hugo de Vries. The visit turned Morgan's interest to questions of heredity and sex determination. Like Wilson, Morgan was a product of American biological training, having received his doctoral education in embryology at Johns Hopkins. Graduating in 1890, he became a professor at Bryn Mawr College, where he worked with Stevens and Wilson, and in 1904 he followed Wilson to Columbia. He would later develop the fundamentals of classical genetics and chromosome theory in his famous *Drosophila* labs with his students Alfred Sturtevant, Calvin Bridges, and Hermann Muller.

To Morgan's mind, Mendelism was a narrow and reductive theory that could not possibly account for the richness, complexity, and wondrous diversity of the natural world. His 1910 article, "Chromosomes and Heredity," famously castigated Mendelians for offering "intellectual security" at the expense of "restlessness of spirit," "adventurous character," and "the modern spirit of scientific theory."²⁹ The theory of inherited factors residing on chromosomes was repugnant to Morgan's training as an embryologist and his inclination toward developmentalist approaches to biology. As he wrote, "Mendelian principles seem to imply that hereditary elements are preformed in the germ cells, and hence exist as small units corresponding to given adult characters." To Morgan, this preformationism seemed to "ignore all that we know about the physiology of development."³⁰

The sex chromosome took the brunt of Morgan's attack. Like Montgomery, Morgan believed that sex is a spectrum trait, exemplified by the cases of gynandromorphs and hermaphrodites.³¹ As Morgan wrote, the model of genetic sex determination proposed by Mendelians such as Bateson, R. C. Punnett, and William Castle (he did not mention Stevens)

presumed an overly binary conception of sex: “[It] assumes a *unit character* that determines femaleness, but for this very reason it introduces an interpretation of sex that is extremely hypothetical. For, the unit character is no longer simply a quantitative factor but a special element that has the power of turning maleness into femaleness. It is an entirely imaginary factor and lacks observational evidence in its support.”³²

The matter of the sex chromosome so troubled Morgan that even after he later assented to Mendelism and the chromosome theory, he remained irked by the excesses of the notion of a chromosome for sex. He spent a hefty five-page digression of the otherwise concise 1915 text, *Mechanism of Mendelian Heredity*, his powerful initial statement of Mendelian genetics and the chromosome theory, formulating an alternative terminology. Avoiding the term “sex chromosome,” he referred instead to the “sex factors on the chromosomes.” He proposed that instead of notating sex as XX and XY—implying that sex is determined by the whole chromosome—researchers ought to employ an alternative notation representing dominant and recessive factors for sex: FF/FO or FmFm/Fmfm (F standing for female-determining factor). Use of the X and Y to represent sex, he argued, should always be qualified so as not to imply chromosomal determination of sex. “[It is only acceptable] to use the symbols for the sex chromosomes as the symbols for the sex factors also, if it is at the same time recognized that the whole chromosome is not involved in determining sex.”³³ The term “sex chromosome,” Morgan wrote, “prejudices” thinking:

There is no need to assume that X is the *sex chromosome* in the sense of carrying sex. The use of this term, I fear, prejudices the situation by the very aptness of the application. It may be that X only means more X, and that this is a factor in sex determination. . . . I should also object . . . on the general grounds that it refers a particular character to a single chromosome.³⁴

Morgan raised the notion that the X is “female determining” as an example. Since males also have an X, as Morgan correctly pointed out, “a ‘female-producing’ X-sperm will give rise to a male if it enters an egg without an X.” Sperm that “are sometimes loosely referred to as male and female,” Morgan noted, “are properly only male-producing and female-producing in certain *combinations*.”³⁵ (Morgan was prescient: chapter 6 of this book explores the persistence of the assumption that the X is the “female chromosome” even today.)

Morgan warned that the so-called sex chromosomes are only “the *best*

known or most visual factor-difference among any number of possible theoretical ones.” He bemoaned that “the identification of sex as a character with the factor for sex determination has led to needless confusion.”³⁶ Maleness and femaleness, Morgan argued, involve many genetic and developmental factors, and “there are factors in the sex chromosomes that affect many parts of the body.”³⁷ He concluded with a flourish: “To assume that all the factors that are shown by the male or by the female must be carried by a sex chromosome of some kind, if carried at all by the chromosomes, is a travesty.”³⁸

THE TRIUMPH OF THE SEX CHROMOSOME

Today, “sex chromosome” is the textbook term for the X and Y, thought of as entirely uncontroversial. How did the “sex chromosome” move from being the derided and unsettled term it once was to the consensus term of today?

One view is that the diverse alternative names for the X and Y—“heterochromosome,” “accessory chromosome,” and “idiochromosome” among them—were fuzzy terms representing early ignorance about the significance and function of the X and Y. Once the chromosomal determination of sex was proved *true*, leading scientists were forced to concede the aptness of the term “sex chromosome.” However, historical evidence does not support this explanation. The eventual acceptance of the notion of a sex chromosome did not follow from empirical confirmation of the theory of the chromosomal determination of sex. While sex chromosomes were shown to *mark* sex and to segregate differently by sex during *gamete formation*, they were never proven to *determine* sex. Well into the 1950s, and some might say even up to today, the precise causal relation between the X, Y, and sex in mammals remained indeterminate. Moreover, it was not obvious that “sex chromosome” was the best technical term for the X and the Y. Alternatives such as “accessory chromosome” and “heterochromosome” were no less precise, generalizable, or descriptively rich than “sex chromosome.” As we have seen, competing terms picked out different features of X and Y structure, behavior, and function that are just as significant as their connection to sex—features that would be sidelined by the eventual triumph of the sex chromosome concept.

To see how the sex chromosome triumphed despite these objections, we must first momentarily step back in our chronology, to the beginnings of the chromosomal theory of inheritance. When the X and Y were first discovered at the turn of the twentieth century, the question of sex

determination was of marginal interest to chromosome scientists. These scientists were seeking to understand the basics of cell biology. What were the chromosomes? How does cell division occur? Where is the hereditary material housed in the cell? The X and Y intrigued cell biologists principally because of their ease of identification and their curious behavior during cell division. In the early years of the twentieth century, these special features helped to reveal a new and broadly explanatory physical theory of heredity known today as the chromosomal theory of inheritance. Sex was both central and incidental to this effort. “Maleness” and “femaleness,” like Mendel’s smooth and wrinkled peas, were salient biological characters used to test, confirm, and evangelize a new theory of inheritance, physically instantiated in the squiggly, paired bodies in the nucleus of the cell. The “sex chromosomes,” like Mendel’s peas, became the mnemonic touchstones of the new chromosome theory. As we shall shortly see, the X chromosome uniquely demonstrated that chromosomes were autonomous units that maintained their constancy in the organism—a necessary feature for any physical carrier of Mendelian traits. Key elements of the chromosomal theory of inheritance emerged from observations of the X and Y chromosomes in the first decades of the twentieth century, including that chromosomes are individual units, pair as homologues, carry hereditary material, and specialize in different genetic factors. These advances brought Mendel’s laws into line with cytology, embryology, and developmental biology, laying the theoretical groundwork for classical genetics.³⁹

THE CHROMOSOMAL THEORY OF INHERITANCE

Walter Sutton, a graduate student of Wilson’s at Columbia, seized on McClung’s demonstration of the individuality and constancy of the X chromosome in 1899 and 1902 to argue forcefully that chromosomes are the units or carriers of heredity (see fig. 3.1).⁴⁰ The X, he argued, showed that chromosomes have characteristic sizes and structures, that they maintain their individuality throughout cell division, and that they carry particular hereditary traits. As Sutton wrote:

Perhaps the most important thing to be gained at present from the knowledge of the behavior of the accessory chromosome . . . is the light which it throws upon the question of the individuality of the chromosomes. . . . Now, if it be admitted that the body *is* a chro-

mosome, inspection quickly shows us that it maintains throughout the spermatogonial divisions, as well as in those that follow, an indubitable independence. . . . Having, then, one of the chromosomes which preserves its individuality in this way, and seeing the other chromosomes enclosed for a part of their development in similar individual vesicles . . . have we not a right to suppose that at one time they too enjoyed the same independence as their more exclusive mate? . . . If this be granted, then we have at least one more ground for belief in the individuality of the chromosomes. . . .⁴¹

In Sutton's 1902 and 1903 work establishing the chromosomal theory of inheritance, the X chromosome featured as the "most unequivocal evidence of chromosomal individuality."⁴² The clincher, of course, was that the X was also associated with the dramatic biological trait of sex. As Sutton wrote, "this particular chromosome" possesses "the power of impressing on the containing cell the stamp of maleness, in accordance with McClung's hypotheses."⁴³ The elegant meaning of Sutton's conclusion, even as it reiterated the erroneous idea that the extra X must be male-determining, could not be missed. In the X chromosome, Sutton had located every piece of evidence required to demonstrate that the chromosomes carry the hereditary material, behave as Mendel predicted, and carry discrete traits with stunning consequences for organismic development.

The role played by the X and Y in the development of the chromosomal theory of inheritance illuminates the "sex chromosomes" not merely as agents of sex determination but as conceptual innovations carved into the fibers and foundations of modern genetics. Historians of biology have not ignored the X chromosome's pivotal role in the development and dissemination of the chromosome theory of heredity. The flip side, however, remains to be considered. What was the role of the chromosome theory of heredity in shaping the "sex chromosome"?

One answer to this question can be found in the remarkable record left by Wilson. In 1905, Wilson recounted how observations of the X and Y ultimately persuaded him of the chromosome theory's validity. As he wrote:

I must frankly confess that until I had followed step by step the behavior of the idiochromosomes . . . I did not appreciate how cogent is the argument . . . that synapsis involves an actual conjugation of chromosomes two by two, and that the chromosomes thus uniting are the paternal and maternal homologues.⁴⁴

Wilson's account points to how consolidating a chromosomal link to the trait of sex served key evangelistic purposes within biology during the first quarter of the twentieth century. Well into the 1910s, much later than is generally appreciated, the sex chromosomes constituted the principal direct evidence for the chromosomal theory of inheritance. The chromosome theory had not gained the adherence of all geneticists—let alone biologists in general—and the sex chromosome remained a central selling point for chromosome theory proponents. As Wilson observed in 1914, “some eminent students of genetics are still reluctant to accept this theory.”⁴⁵ He continued, “the conjugation of chromosomes, to say nothing of paternal and maternal homologues, has been obstinately contested; it must be admitted that the proof is still far from complete for the chromosomes generally,” however, “in the case of the sex-chromosomes . . . the probability becomes a certainty.”⁴⁶

Wilson's 1914 lecture to the Royal Society of London, “The Bearing of Cytological Research on Heredity,” trotted out the sex chromosomes to illustrate every major insight of chromosome research in the post-Mendelian era, including independent assortment, conjugation, and linkage. Wilson wrote:

To the cytologist the *interest of the phenomena extends far beyond the special problem of sex*. Nature has here performed a series of experiments which gives a crucial test of many of our earlier conclusions, provides a secure basis for further advances, and at the same time brings vividly before us the connection of the chromosomes with heredity.⁴⁷

He continued:

. . . The cytological phenomena of sex-production lend strong support to the theory of the genetic continuity of chromosomes. They give unquestionable proof, in the case of a particular chromosome pair (XY), of the conjugation and subsequent disjunction of corresponding maternal and paternal chromosomes. . . . They give the first direct evidence of a difference of nuclear constitution between the homozygous and the heterozygous conditions, and of corresponding gametic differences. And finally . . . they fully substantiate the general cytological explanation that has been offered of Mendel's law.⁴⁸

The sex chromosome, therefore, was not only an object of intense interest and debate, from a broad range of angles within genetic theory, during the critical period of the development of classical genetics. It was also a simple explanatory hook with which one could evangelize for the chromosomal theory of inheritance. It certainly did not hurt that the chromosomal trait was the dramatic and tangible phenotypic trait of sex, an object of popular fascination and a long-standing target of biological investigation.

MORGAN'S SEX-LINKED TRAITS

The story of the role that the chromosomal theory of inheritance played in establishing the “sex chromosome” continues in Morgan's fly lab in the years 1910 to 1920. Most flies have red eyes, but in 1910 Morgan found a rare mutant white-eyed male. When crossed with a red-eyed female, white-eyed males disappeared for a generation but then showed up in the subsequent one. When a white-eyed female was crossed with a red-eyed male, all of the male progeny had white eyes. The only explanation for this pattern of inheritance, now familiar to us from the case of human hemophilia, was that the white-eyed variant was recessive and carried on the X chromosome. It was “sex linked.”⁴⁹

Sex linkage allowed normal phenotypic traits to be mapped to the X chromosome and opened up a world of chromosomal linkage mapping. Morgan subsequently developed sex linkage into an experimental system and chromosome-mapping industry for genetics. All of the first genetic mutations, and many of the most important principles of genetic inheritance, were found on the X chromosome. Over the next ten years, with the aid of the X chromosome and the fruit fly's simple biology, short lifespan, and four plump chromosome pairs, Morgan and his students elaborated the chromosomal basis of inheritance (see fig. 3.2). Morgan's detailed studies of sex linkage in *Drosophila* ultimately provided the first definitive experimental evidence for the alignment of Mendelism with the chromosome theory of heredity.

Morgan, we recall, had vigorously objected to the notion of a “chromosome for sex.” He railed at length against its imprecision and the view of heredity, development, and life that it seemed to represent. He was an antichromosome theory partisan, and the sex chromosomes represented everything that was wrong with the theory. The concept of the sex chromosome, he wrote, was a “travesty.” Yet in 1910, after a decade of

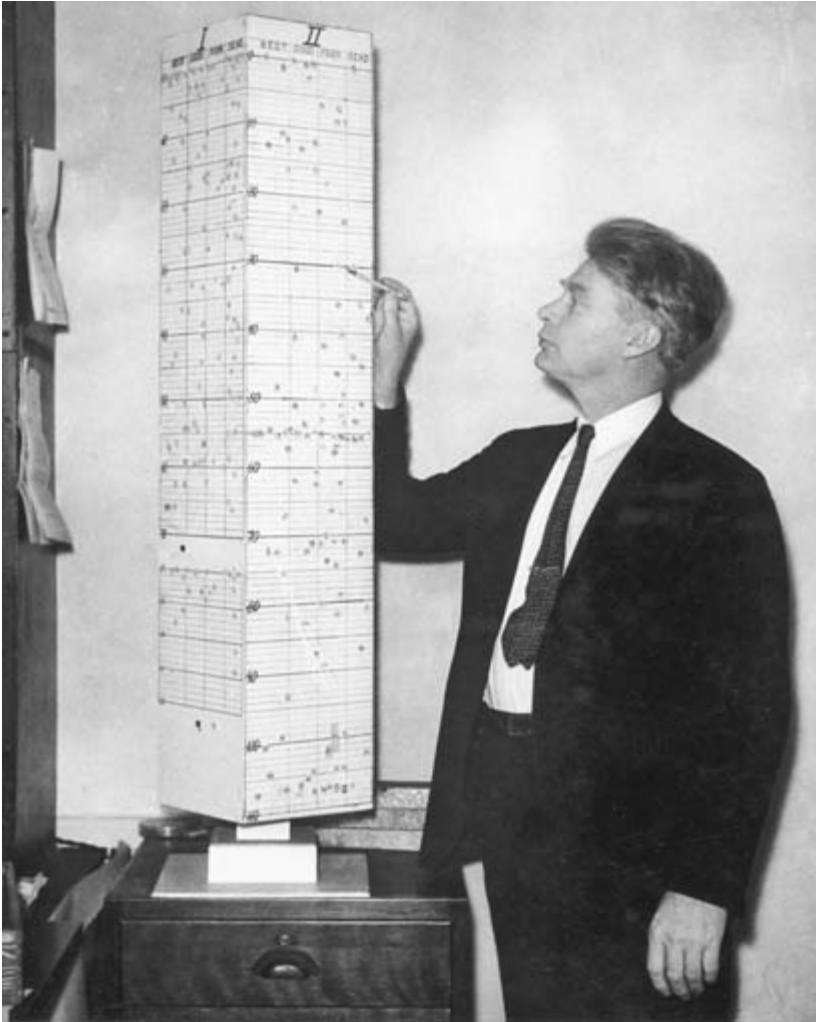


Figure 3.2. Calvin Bridges mapping genes to the *Drosophila* chromosomes. Courtesy of the Archives, California Institute of Technology.

principled resistance, Morgan began to relent. In short order, he became the principal champion of Mendelism and the chromosome theory. By 1916, he had even adopted the term “sex chromosome” in his own writing. His shift had everything to do with the special properties of the X and Y chromosomes.

Morgan’s change of view began with his acceptance, in 1910, of evidence from Nettie Stevens and Edmund Wilson, about the role of the

X and Y in sex determination: that there are “two classes of sperm, differentiated on the basis of the accessory chromosome,” that in meiosis a separation of male and female elements does occur, and that Wilson’s and Stevens’s model held for higher animals as well as plants and insects.⁵⁰ This was persuasive evidence, indeed, that chromosomes might carry hereditary elements. The most important factor in Morgan’s shift in perspective, though, was his own finding of his famous white-eyed male in *Drosophila*: the first sex-linked genetic mutation.

As historian of science Sharon Kingsland writes, Morgan’s 1913 *Heredity and Sex* “cast the entire problem of sex determination in a Mendelian framework.”⁵¹ Morgan now championed the chromosome theory against those voicing his own previous doubts. Whereas in 1910 Morgan decried the chromosome theory as “violating the modern spirit of scientific theory,” in his 1915 *The Mechanism of Mendelian Heredity* he wrote, “why then, we are often asked, do you drag in the chromosomes? Our answer is that since the chromosomes furnish exactly the kind of mechanism that the Mendelian laws call for; and since there is an ever-increasing body of information that points clearly to the chromosomes as bearers of the Mendelian factors, it would be folly to close one’s eyes to so patent a relation.”⁵² Whereas in 1915 Morgan was still denouncing the misleading nature of the term “sex chromosome,” in his 1916 *Sex-Linked Inheritance in Drosophila*, Morgan acceded, writing with Bridges that, “following Wilson’s nomenclature, we speak of both X and Y as sex chromosomes.”⁵³



Critics had attacked the sex chromosome for its simplistic binary conception of sex, for its failure to explain sex in all species, and for lacking a mechanism for its role in determining sex. The notion of a chromosome for sex survived these attacks, in part due to its intimate and indispensable role in building the case for the new chromosomal theory of heredity. The explanatory success of the chromosomal theory of heredity ultimately overwhelmed critics’ worries about the sex chromosome. These critics silenced their objections, not because they were satisfactorily addressed (even after he accepted Mendelism, Morgan still did not feel the matter of the sex chromosome was resolved, as we can see from his 1915 discourse), but because the broader explanatory utility of the chromosomal theory simply swept them aside. In the face of the chromosome theory’s explanatory success, critiques of the sex chromosome became minor and distracting terminological quibbles. The priority of generat-

ing a consensus around the chromosome theory overtook the sex chromosome dispute. The case of Morgan vividly demonstrates this point.

The term and the concept of the “sex chromosome” gathered steam throughout the 1920s, and by the 1930s it was the term of choice for the X and Y chromosomes. Alternative terms, such as “accessory chromosome” and “heterochromosome,” drifted into disuse. Though the term “sex chromosome” had triumphed, few maintained that the X itself was *sex determining*. Wilson was an exception. He maintained that the X and Y were, in one way or another, the material hereditary basis for maleness and femaleness. Wrote Wilson in 1925: “That these chromosomes bear the differential factors of sex has been proved to be the case by an irresistible body of more detailed and concrete evidence drawn from both genetics and cytology. The chromosomes in question may, therefore, appropriately be called the *sex-chromosomes*.”⁵⁴

Others resolved that the X is the “sex chromosome” only in the sense that it serves as a histological *marker* of sex. For instance, as the Edinburgh geneticist F. A. E. Crew wrote in a text on the genetics of sex in 1927, “since the tissues from male and female differ chromosomally only in this respect, these chromosomes are referred to as the sex-chromosomes.” Notably, this formulation altogether avoids the matter of the causal relation between the X and the sex.⁵⁵

Most commonly, the “sex chromosome” construct derived from the phenomenon of *sex linkage*, which lay at the heart of the new genetic research program produced by Morgan and his students. This “sex chromosome” concept entirely sidestepped the question of the X’s role in sex determination. *Drosophila* geneticists were not primarily interested in the question of sex determination, but rather in sex linkage as a clue to traits especially amenable to genetic study. As Morgan and Bridges wrote in the 1916 monograph, *Sex-Linked Inheritance in Drosophila*:

This term (sex-linked) is intended to mean that such characters are carried by the X chromosome. It has been objected that this use of the term implies a knowledge of a factor for sex in the X chromosome to which the other factors in that chromosome are linked . . . [but] we are unable as yet to locate the sex factor or factors in the X chromosome.⁵⁶

In this formulation, the X was a “sex chromosome” not because it was the sex determiner, but because it carried at least one critical factor corresponding to sex, causing traits on the X to be reliably sex linked.

Perhaps unknown to the scientists at the 1960 Denver conference,

then, their questions about the reasoning behind the inherited terminology of the “sex chromosome” were not new. Doubts about the precision and clarity of the concept of the “sex chromosome” have been a persistent theme since the discovery of the X and Y. The X and Y debuted in a charged moment in the history of biology when the fundamentals of genetics and heredity fell into place. They became a centerpiece of this effort. At the same time, they were ambivalently and instrumentally joined to sex, opening a speculative space that carves the terrain of this book.

In the first years of the twentieth century, cytologists asserted that chromosomes carry the hereditary material and that they provide the mechanism for the generation and recombination of novel hereditary characters. McClung’s, Stevens’s, and Wilson’s evidence for an association between chromosomal complement and sex, and the visually compelling case that the X and Y made for the individuality and constancy of the chromosomes, offered the principal early evidence for the chromosome theory. In the 1910s, this theory was further confirmed and extended by studies of X-linked traits. Sex linkage became the principal system for early studies of genetic mutations. The link between the X and sex became the crucial fact in these scientific research programs, and in its powerful simplicity the so-called sex chromosome became the prized plume of the new genetics.

Terms such as “heterochromosome” and “accessory chromosome,” while more consistent with terminological convention, would not do. They did not point to the critical relationship between a chromosome and an obvious phenotypic trait. The clear way in which the X and Y marked sex difference became important: *rhetorically*, in the recruitment of the biological community to the new theory, and *scientifically*, as the central technology of the “sex-linkage” research program that confirmed the chromosomal basis of Mendelian traits. For these aims, terms that were perhaps more descriptive, capturing the structural and behavioral features of the X and Y, went by the wayside. Wilson’s efficient shorthand of the “sex chromosome” won the day.

